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## ABSTRACT

**Background.** ICU mortality prediction models trained on electronic health records face pervasive Missing-Not-At-Random (MNAR) data caused by selective clinical test ordering. Standard gradient boosting methods achieve competitive discrimination within their training distribution by treating missingness as a proxy variable, but degrade substantially when missingness patterns shift between institutions. No existing framework simultaneously quantifies imputation uncertainty, propagates it structurally to a downstream classifier, and converts it into principled prediction abstention.

**Methods.** We introduce JUCO (Joint Uncertainty-Calibrated Optimisation), a four-phase framework. LightGBM quantile regression produces interval-valued imputations  $[q^L, q^U]$  rather than point estimates. Intervals are encoded as trapezoidal fuzzy numbers whose support width represents imputation uncertainty. A fuzzy decision forest soft-routes samples, down-weighting imputed inputs in leaf votes. An entropy-based safe abstention protocol defers high-uncertainty predictions to clinician review. All five hyperparameters are jointly optimised via two-stage Differential Evolution on MIMIC-IV, then frozen for external validation on eICU (208 hospitals).

**Results.** Under escalating MNAR stress (10%–70%), JUCO's AUROC degrades by  $-0.023$  versus  $-0.054$  for XGBoost and  $-0.056$  for MissForest+XGB; JUCO overtakes both at 70% missingness (0.799 vs 0.795 and 0.792). Safe abstention yields a  $2.8\times$  AURC reduction versus full-coverage prediction (0.032 vs 0.090). Under standard 30% MNAR, the AUROC gap between gradient boosting and JUCO narrows by 46% from MIMIC-IV to eICU ( $-0.028$  to  $-0.015$ ), consistent with missingness exploitation diminishing under cross-institutional shift.

**Conclusions.** JUCO provides principled uncertainty propagation from missing data to clinical prediction deferral, offering the greatest advantage in environments where data pipeline integrity cannot be guaranteed. Code available at [https://github.com/\[REDACTED-BEFORE-SUBMISSION\]](https://github.com/[REDACTED-BEFORE-SUBMISSION]).

## 1. Introduction

Accurate prediction of in-hospital mortality for ICU patients enables early intervention, targeted resource allocation, and timely initiation of goals-of-care conversations. Over the past decade, machine learning models trained on electronic health records (EHR) — particularly gradient boosting methods — have achieved discrimination levels comparable to established clinical scores such as APACHE and SOFA (Purushotham et al., 2018; Johnson et al., 2023), with AUROC values routinely in the range 0.82–0.88 on retrospective cohorts. Yet deployment of these models in real ICU environments remains limited (Topol, 2019), and a persistent obstacle is the pervasive, structured missingness of EHR data. A recent meta-analysis of 40 ICU mortality prediction studies confirms gradient boosting as the dominant modality, yet identifies systematic calibration deficits as the primary unresolved limitation (Sun et al., 2025a).

ICU laboratory and vital sign measurements are not missing at random. A clinician's decision to order a lactate measurement, a blood culture, or an arterial blood gas is itself a clinical judgment driven by patient acuity — the probability that a value is measured is correlated with the value itself and with the patient's probability of death. This Missing-Not-At-Random (MNAR) mechanism (Rubin, 1976) means that the *absence* of a measurement carries prognostic information, and that the distribution of missing data at test time may differ from training time as hospital protocols, equipment, staffing, and patient populations evolve. Benchmark studies confirm that clinical risk models degrade systematically when evaluated on external populations (Moons et al., 2012), and recent work attributes a significant fraction of this degradation to unstable missingness patterns (Agniel et al., 2018; Josse et al., 2019).

Existing approaches to missing data in clinical ML occupy two extremes that both fail to address this challenge fully. *Point imputation methods* — mean imputation, MICE, MissForest — produce single values for each missing entry and discard uncertainty, giving downstream classifiers no mechanism to distinguish confidently observed features from highly uncertain imputed ones. *Native NaN routing* in gradient boosting methods (XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017)) avoids imputation entirely but encodes the MNAR structure as a proxy variable whose stability depends on the stationarity of the missingness mechanism between training and deployment. Neither approach quantifies imputation

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uncertainty, neither propagates it structurally to the classifier, and neither provides a mechanism for identifying and deferring predictions when imputation quality is low. In safety-critical deployment, this is a failure mode: a model that issues a confident mortality prediction for a patient whose key laboratory values are all imputed provides false certainty that can harm rather than help clinical decision-making.

We introduce **JUCO** (Joint Uncertainty-Calibrated Optimisation), a four-phase framework that treats imputation uncertainty as a first-class quantity throughout the prediction pipeline. The key insight is that a quantile regression interval  $[q^L, q^U]$  directly encodes imputation confidence: a wide interval indicates low confidence, a narrow interval high confidence. Encoding this interval as a trapezoidal fuzzy number and routing samples *softly* through a decision tree — so uncertain samples split their weight across both children in proportion to their uncertainty — propagates imputation quality to the tree's leaf distribution. The resulting entropy of the leaf distribution then drives abstention: high entropy triggers deferral to clinical review. This chain — from missing data to interval to fuzzy number to soft routing to entropy to abstention — is end-to-end, structurally motivated, and jointly optimised via Differential Evolution.

We evaluate JUCO on MIMIC-IV (Johnson et al., 2023) (82,785 ICU stays, Beth Israel Deaconess Medical Center) as a primary cohort and on the eICU Collaborative Research Database (Pollard et al., 2018) (162,136 stays across 208 US hospitals) as an external validation cohort with parameters frozen from MIMIC-IV training. Under escalating informative missingness (10%–70%), JUCO's AUROC degrades by  $-0.023$ , compared to  $-0.054$  for XGBoost and  $-0.056$  for MissForest+XGB; JUCO overtakes both models at 70% missingness (0.799 vs 0.795 and 0.792). The safe abstention protocol reduces AURC by  $2.8\times$  versus full-coverage prediction (0.032 vs 0.090). Under standard 30% MNAR, gradient boosting achieves statistically significantly higher AUROC than JUCO (0.842 vs 0.814, DeLong  $z = -18.04$ ,  $p < 0.001$ ); this gap narrows by 46% on eICU ( $-0.028$  MIMIC-IV vs  $-0.015$  eICU), consistent with gradient boosting's missingness-exploitation advantage diminishing under cross-institutional distributional shift.

The specific contributions of this paper are:

1. A four-phase framework that structurally propagates quantile-regression imputation uncertainty through a fuzzy decision forest to entropy-based clinical abstention, jointly optimised end-to-end via two-stage Differential Evolution.
2. A systematic missingness stress test across seven MNAR rates (10%–70%) demonstrating that JUCO's graceful degradation ( $\Delta\text{AUROC} = -0.023$ ) is substantially smaller than gradient boosting baselines ( $-0.054$  to  $-0.056$ ).
3. An ablation study isolating each architectural component: the safe abstention protocol ( $2.8\times$  AURC improvement) and fuzzy soft routing (12.7% ECE reduction vs crisp routing).
4. Transparent reporting of JUCO's lower nominal AUROC under standard conditions with a mechanistic explanation based on gradient boosting's use of missingness as a stationary proxy variable, and empirical evidence that this advantage diminishes under distributional shift.
5. Cross-institutional external validation on eICU with frozen parameters demonstrating a 46% reduction in the AUROC gap.

The remainder of this paper is organised as follows. Section 2 surveys related work. Section 4 formalises the problem. Section 5 describes the JUCO architecture. Section 6 describes datasets, protocol, and baselines. Section 7 presents experimental results. Section 8 discusses findings and limitations. Section 9 concludes. This study follows the TRIPOD+AI reporting guidelines for clinical prediction models (Collins et al., 2024).

## 2. Related Work

This section surveys four bodies of literature that directly motivate and situate the contributions of this paper: missing data in EHR-based clinical prediction, imputation under informative missingness, fuzzy set methods for uncertain data, and selective prediction with abstention.

### 2.1. Missing Data in ICU Clinical Prediction

Electronic health records contain pervasive missingness by design: clinicians order tests selectively based on patient acuity, meaning the decision to measure a variable is correlated with the variable's value and the patient's outcome (Agniel et al., 2018). This produces a Missing-Not-At-Random (MNAR) mechanism in the formal sense of Rubin (1976): the probability that an entry is missing depends on the missing value itself, not only on observed data. Standard missing-at-random (MAR) or missing-completely-at-random (MCAR) assumptions that underpin most imputation methods are therefore violated in ICU data, with consequences for both imputation quality and downstream model validity.

Benchmark studies on MIMIC-III/IV have demonstrated that gradient boosting methods — particularly XGBoost and LightGBM — achieve competitive ICU mortality prediction performance (Johnson et al., 2016, 2023; Purushotham et al.,

2018), with AUROC values in the range 0.82–0.88. A recent systematic review and meta-analysis of 40 ICU mortality prediction studies confirms gradient boosting as the dominant modality, with a pooled AUROC of 0.83, yet identifies systematic calibration deficits as the primary unresolved limitation across all model classes (Sun et al., 2025b). Complementary work on myocardial infarction patients using eICU and MIMIC-IV demonstrates that explainable XGBoost frameworks can achieve AUROC values up to 0.92 with time-resolved interpretability across multiple prediction horizons (Mesinovic et al., 2025), while a dynamic LightGBM model updated every 24 hours achieves an internal AUROC of 0.909 with successful external validation on MIMIC-IV (Britsch et al., 2025). These methods handle missingness implicitly through learned NaN routing: split directions for missing entries are determined during training to minimise loss, effectively treating the absence of a measurement as a learnable feature (Chen and Guestrin, 2016). While this approach is efficient and often effective within a single institution’s stable data pipeline (Perez-Lebel et al., 2022), it encodes the missingness structure as a proxy variable whose validity depends on the stationarity of the MNAR mechanism (Josse et al., 2019; Perez-Lebel et al., 2022). When the deployment environment differs from training — different hospital charting protocols, data pipeline failures, or equipment changes — the learned NaN routing becomes miscalibrated (Sperrin and Martin, 2020). Our stress test results (XGBoost  $\Delta$ AUROC =  $-0.054$  at 70% MNAR vs JUCO  $-0.023$ ) are consistent with this fragility.

Reliable external validation of clinical prediction models requires careful attention to study design and statistical power. Riley et al. (2024) demonstrate that most published external validation studies are underpowered to detect clinically meaningful performance degradation, and advocate for setting-specific sample size calculations rather than rules of thumb. We follow this guidance in our eICU validation design, which applies frozen MIMIC-IV parameters to 162,136 stays across 208 hospitals.

## 2.2. Imputation Methods for Informative Missingness

Standard imputation approaches for clinical data include mean/median imputation (Sterne et al., 2009), multiple imputation by chained equations (MICE (van Buuren and Groothuis-Oudshoorn, 2011)), and MissForest (Stekhoven and Bühlmann, 2012). All of these produce point estimates and discard uncertainty information, leaving the downstream classifier unable to distinguish between confidently imputed and highly uncertain entries. A systematic review of missing-data handling across 46 EHR studies finds that machine learning methods generally outperform statistical approaches, but that no single strategy is universally optimal: MICE-based methods are preferable for cross-sectional data, while KNN and context-aware approaches perform better for longitudinal records (Ren et al., 2024). A broader survey of 58 clinical studies on imputation for structured datasets confirms that the choice of method should account for missingness mechanism, pattern, and ratio, rather than being selected by default (Afkanpour et al., 2024).

Deep generative approaches address this limitation by modelling the full conditional distribution of missing values. Yoon et al. (2018) propose GAIN, a GAN-based imputation method. Ipsen et al. (2021) extend variational autoencoders to MNAR data, producing distributional imputations. Ma et al. (2020) use recurrent networks with explicit missingness modelling for temporal EHR data. More recently, Zhao et al. (2024) propose a multi-task diffusion model for mortality prediction with incomplete data in shock patients. A comprehensive benchmarking study of deep imputation methods across five health time-series datasets finds that no single method dominates, and that deep learning outperforms traditional approaches only when data quality and sample size are sufficiently large (Kazijevs and Samad, 2023). A parallel review comparing GAIN, VAE, MICE, and MissForest under MCAR, MAR, and MNAR conditions on ten tabular datasets concludes that conventional methods are preferable for moderate sample sizes under MNAR, with deep generative methods prone to mode collapse in that setting (Sun et al., 2023). These approaches produce richer uncertainty representations, but they do not propagate imputation uncertainty to the classification stage and do not incorporate abstention.

The handling of missing data specifically at model deployment time introduces additional challenges beyond those encountered during development. Sisk et al. (2023) show via simulation on critical-care data that missing indicators improve predictive performance in many scenarios but can introduce bias under outcome-dependent MNAR mechanisms, motivating principled uncertainty representations such as JUCO’s interval-valued imputation over binary missingness flags.

Quantile regression provides an efficient alternative for interval-valued imputation (Meinshausen, 2006). The key advantage is that prediction interval width directly encodes imputation uncertainty without distributional assumptions. To our knowledge, the present work is the first to use LightGBM quantile regression as an imputation engine whose output intervals are propagated through a downstream soft-routing classifier via a fuzzy transformation.

## 2.3. Fuzzy Set Methods for Uncertain Clinical Data

Fuzzy set theory (Zadeh, 1965) provides a principled framework for reasoning about imprecise information. Trapezoidal fuzzy numbers have been applied to clinical decision support for representing measurement uncertainty (Adlassnig, 1986; Torres and Nieto, 2013), but these applications use fuzzy representations at the rule level rather than as inputs to a learned classifier. Fuzzy decision trees with soft routing have been studied theoretically (Olaru and Wehenkel, 2003; Yuan and Shaw, 1995), but prior work has not addressed joint optimisation of imputation and tree hyperparameters, nor application to large-scale EHR data. Hüllermeier and Waegeman (2021) motivates the distinction between aleatoric and epistemic uncertainty that guides our design choices.

More recently, [Capitoli et al. \(2025\)](#) propose Fuzzy Sets in Probability Trees (FPT), which embed fuzzy membership functions within probabilistic decision trees to preserve nuances lost when crisp thresholds are applied. Applied to thyroid nodule classification and chronic kidney disease progression, FPTs provide interpretable uncertainty-aware clinical decision support that reduces misdiagnosis rates relative to standard decision trees. JUCO extends this principle to large-scale EHR data under MNAR missingness, replacing rule-level fuzzy logic with imputation-interval-driven soft routing jointly optimised end-to-end via Differential Evolution.

## 2.4. Selective Prediction and Safe Abstention

Selective prediction — the reject option — allows a model to withhold predictions for uncertain inputs ([Chow, 1957](#); [Bartlett and Wegkamp, 2008](#)). [Geifman and El-Yaniv \(2017\)](#) establish risk-coverage guarantees for selective prediction and introduce the AURC metric used in this paper. [Franc et al. \(2023\)](#) show that entropy is asymptotically optimal as an abstention criterion under a well-calibrated posterior.

In clinical ML, [Jones et al. \(2021\)](#) demonstrate entropy-based abstention for risk stratification, and [Kompa et al. \(2021\)](#) show AURC improvements across multiple clinical datasets. [Leibig et al. \(2017\)](#) demonstrate that deferring uncertain predictions improves system-level accuracy in diabetic retinopathy screening. More recently, [Swaminathan et al. \(2024\)](#) show that selective prediction reduces total misclassification cost in structured EHR data extraction tasks, demonstrating broad applicability of abstention in clinical ML pipelines.

A growing body of work extends selective prediction to handle distributional shift between training and deployment. [Kwon and Kim \(2026\)](#) propose a conformal selective prediction framework for clinical triage that combines conformal prediction with cost-aware deferral; on an early sepsis task, selective deferral reduces error by 46.7% at 80% coverage even under out-of-distribution temporal shift, a result structurally analogous to JUCO's entropy-driven abstention under MNAR shift. [Hasan et al. \(2023\)](#) provide a systematic survey of the reject option in neural networks, cataloguing loss functions and post-training strategies for knowledge-aware abstention.

The key limitation of all prior clinical selective prediction work is that abstention is based on classification boundary uncertainty — not on uncertainty in the input features themselves. JUCO addresses this gap: wide imputation intervals produce wide fuzzy trapezoids, reduce split sharpness (Eq. 13), produce diffuse leaf distributions, and increase output entropy. The abstention decision is mechanistically linked to imputation quality, not merely to the classification margin.

## 3. Related Works

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## 4. Problem Formulation

### 4.1. Clinical Setting and Notation

Let  $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{M}_i, y_i)\}_{i=1}^n$  denote a dataset of  $n$  ICU stays, where:

- $\mathbf{x}_i \in (\mathbb{R} \cup \{\text{NaN}\})^d$  is the feature vector of  $d$  clinical variables aggregated over the first 24 hours of admission;
- $\mathbf{M}_i \in \{0, 1\}^d$  is the missingness indicator vector, with  $M_{ij} = 1$  iff feature  $j$  is missing;
- $y_i \in \{0, 1\}$  is the binary in-hospital mortality label.

Table 1 summarises all symbols used in this paper.

### 4.2. The Informative Missingness Problem

Following Rubin (1976), the missingness mechanism is characterised by:

$$P(\mathbf{M} \mid \mathbf{X}_{\text{full}}, \mathbf{y}). \quad (1)$$

In ICU data, this distribution is not ignorable. A laboratory value such as lactate is ordered when the attending clinician suspects tissue hypoperfusion — the physiological state most associated with high mortality. This creates the MNAR condition:

$$P(M_{ij} = 1 \mid \mathbf{x}_{ij}^{\text{true}}, \mathbf{x}_{i,-j}, y_i) \neq P(M_{ij} = 1 \mid \mathbf{x}_{i,-j}, y_i). \quad (2)$$

Table 1: Summary of notation.

| Symbol                                                         | Definition                                   |
|----------------------------------------------------------------|----------------------------------------------|
| $\mathbf{X} \in (\mathbb{R} \cup \{\text{NaN}\})^{n \times d}$ | Feature matrix with informative missingness  |
| $\mathbf{M} \in \{0, 1\}^{n \times d}$                         | Missingness mask ( $M_{ij} = 1$ iff missing) |
| $\mathbf{y} \in \{0, 1\}^n$                                    | Binary mortality labels                      |
| $\alpha_L \in (0, 0.5)$                                        | Lower quantile level                         |
| $\alpha_U \in (0.5, 1)$                                        | Upper quantile level                         |
| $q_{ij}^L, q_{ij}^M, q_{ij}^U$                                 | Imputed lower, median, upper bounds          |
| $\eta \in [0, 0.5]$                                            | Fuzzy proportion factor                      |
| $(a_{ij}, b_{ij}, c_{ij}, d_{ij})$                             | Trapezoidal fuzzy number for $(i, j)$        |
| $\theta = (\alpha_L, \alpha_U, \eta, d_{\max}, n_{\min})$      | DE-optimised parameter vector                |
| $w_k$                                                          | Inverse-frequency class weight               |
| $H_i$                                                          | Shannon entropy of predicted probabilities   |
| $\tau$                                                         | Entropy abstention threshold                 |
| PICP                                                           | Prediction Interval Coverage Probability     |
| MPIW <sub>norm</sub>                                           | Normalised Mean Prediction Interval Width    |
| AURC                                                           | Area Under the Risk-Coverage curve           |
| Risk@90                                                        | Error rate on the most confident 90%         |

### 4.3. Formal Problem Statement

**Problem 1 (Uncertainty-Aware Mortality Prediction Under MNAR).** Given training data  $D_{\text{train}}$  drawn from a distribution satisfying Eq. 2, learn a selective prediction function:

$$f : (\mathbb{R} \cup \{\text{NaN}\})^d \rightarrow [0, 1] \cup \{\perp\}, \quad (3)$$

where  $\hat{p}_i \in [0, 1]$  is a calibrated mortality probability and  $\perp$  indicates abstention, such that: (i) predicted probabilities are well calibrated on retained cases; (ii) abstention is concentrated on high imputation-uncertainty cases; and (iii) AUROC( $f, r$ ) degrades slowly as the additional MNAR rate  $r$  increases.

### 4.4. Evaluation Objectives

Evaluations prioritise the three requirements of Problem 1 in the following order: (1) robustness under missingness shift (stress test, 10%–70%); (2) abstention quality (AURC, ablation); (3) cross-institutional generalisation (eICU, frozen parameters); (4) discrimination under standard conditions (AUROC with explanation); (5) calibration (ECE, reliability diagrams).

## 5. Methodology

JUCO (Joint Uncertainty-Calibrated Optimisation) addresses the limitations described in Section 4 through a four-phase pipeline, illustrated in Figure 1. Each phase is designed so that uncertainty from earlier stages is *preserved* and *amplified* through later stages rather than collapsed to a point estimate: (1) quantile regression preserves imputation uncertainty as an interval; (2) fuzzy transformation encodes the interval width as the fuzzy support; (3) soft routing lets imputed samples influence leaf votes proportionally less than observed samples, while contributing to the entropy of the leaf distribution; (4) entropy-based abstention defers the resulting high-uncertainty predictions to human clinicians. The five parameters governing this chain —  $(\alpha_L, \alpha_U, \eta, d_{\max}, n_{\min})$  — are jointly optimised via two-stage Differential Evolution rather than tuned independently.

### 5.1. Phase 1: Adaptive Quantile Imputation

For each feature  $j \in \{1, \dots, d\}$  with missing entries, we train a LightGBM quantile regressor to predict the conditional quantiles of  $x_{ij}$  given the remaining features  $\mathbf{x}_{i,-j}$ . The quantile regressor minimises the weighted pinball loss:

$$\hat{q}_j^\alpha(\mathbf{x}_{-j}) = \arg \min_f \sum_{i: M_{ij}=0} w_i \rho_\alpha(x_{ij} - f(\mathbf{x}_{i,-j})), \quad (4)$$

where  $\rho_\alpha(u) = u(\alpha - \mathbf{1}_{u < 0})$  is the pinball (check) loss at quantile level  $\alpha$  (Koenker and Bassett, 1978), and  $w_i$  are inverse-frequency sample weights that give rare positive outcomes (ICU mortality) equal aggregate influence as the majority class. Specifically, the weight assigned to each sample from class  $k$  is:

$$w_k = \frac{N}{K \cdot n_k}, \quad (5)$$

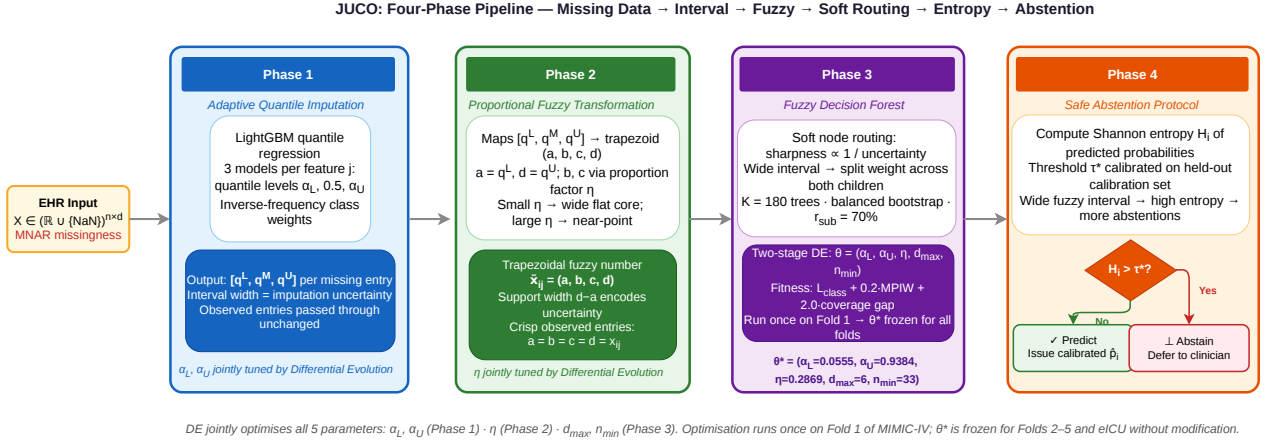


Figure 1: Overview of the JUCO four-phase pipeline. Phase 1 (Adaptive Quantile Imputation) produces prediction intervals  $[q^L, q^U]$  for each missing entry via LightGBM quantile regression. Phase 2 (Proportional Fuzzy Transformation) maps these intervals to trapezoidal fuzzy numbers  $(a, b, c, d)$  whose support width encodes imputation uncertainty. Phase 3 (Fuzzy Decision Forest) routes samples softly through trees, with uncertain (wide-interval) samples distributing their weight across both branches at each node. Phase 4 (Safe Abstention) calibrates an entropy threshold  $\tau$  on a held-out calibration set; test samples with  $H_i > \tau$  are deferred to clinical review.

where  $N$  is the total number of training samples,  $K = 2$  is the number of classes, and  $n_k$  is the count of class- $k$  samples. At  $\approx 10\%$  mortality prevalence ( $n_1 \approx 0.1N$ ), deceased patients receive approximately ten times the weight of surviving patients.

Three regressors are trained per feature at quantile levels  $\alpha \in \{\alpha_L, 0.5, \alpha_U\}$ , producing for each missing entry  $(i, j)$  the triple  $(q_{ij}^L, q_{ij}^M, q_{ij}^U)$  representing the lower bound, median, and upper bound of the imputed interval. The quantile levels  $\alpha_L$  and  $\alpha_U$  are not fixed; they are part of the parameter vector  $\theta$  optimised by Differential Evolution (Section 5.3.5). This allows the framework to adapt the width of imputation intervals to the validation loss, rather than pre-specifying a fixed confidence level.

For *observed* entries ( $M_{ij} = 0$ ), no imputation is performed: the original value  $x_{ij}$  is used directly.

## 5.2. Phase 2: Proportional Fuzzy Transformation

Each imputed interval  $[q_{ij}^L, q_{ij}^M, q_{ij}^U]$  is mapped to a *trapezoidal fuzzy number* (Zimmermann, 2001)  $\tilde{x}_{ij} = (a_{ij}, b_{ij}, c_{ij}, d_{ij})$  with membership function:

$$\mu_{\tilde{x}_{ij}}(x) = \begin{cases} 0 & x < a_{ij} \text{ or } x > d_{ij} \\ \frac{x - a_{ij}}{b_{ij} - a_{ij}} & a_{ij} \leq x < b_{ij} \\ 1 & b_{ij} \leq x \leq c_{ij} \\ \frac{d_{ij} - x}{d_{ij} - c_{ij}} & c_{ij} < x \leq d_{ij} \end{cases} \quad (6)$$

The four parameters are constructed from the quantile outputs as:

$$a_{ij} = q_{ij}^L, \quad (7)$$

$$d_{ij} = q_{ij}^U, \quad (8)$$

$$b_{ij} = \max\left(a_{ij}, q_{ij}^M - \eta \cdot (q_{ij}^U - q_{ij}^L)\right), \quad (9)$$

$$c_{ij} = \min\left(d_{ij}, q_{ij}^M + \eta \cdot (q_{ij}^U - q_{ij}^L)\right), \quad (10)$$

where  $\eta \in [0, 0.5]$  is the *proportion factor* that controls the ratio of the flat core  $[b, c]$  to the total support. Small  $\eta$  yields a wide flat core (interval-like imputation); large  $\eta$  yields a peaked trapezoid (near-point imputation). For *observed* entries the trapezoid degenerates to a crisp value:  $a_{ij} = b_{ij} = c_{ij} = d_{ij} = x_{ij}$ .

Two derived quantities from the trapezoid are used in the routing and optimisation stages:

$$\text{centroid}(\tilde{x}_{ij}) = \frac{a_{ij} + b_{ij} + c_{ij} + d_{ij}}{4}, \quad (11)$$

$$\text{uncertainty}(\tilde{x}_{ij}) = d_{ij} - a_{ij}. \quad (12)$$

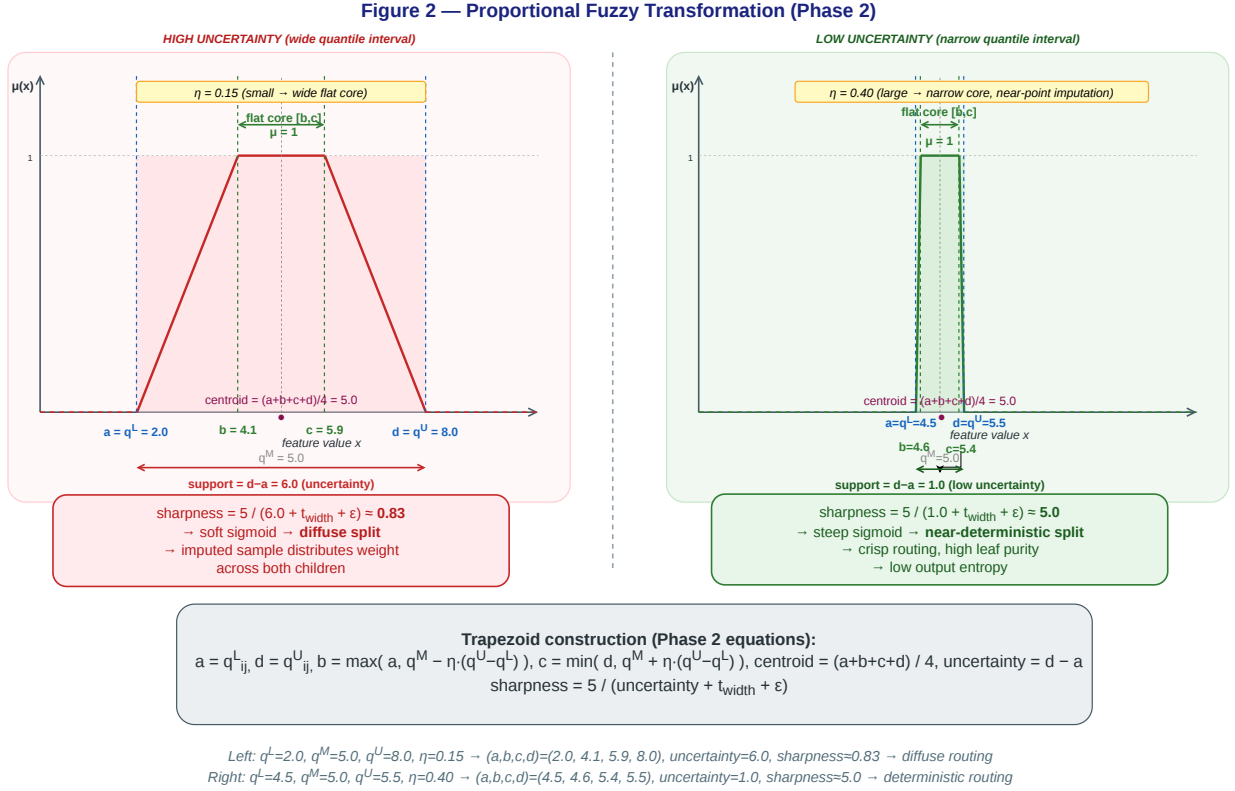


Figure 2: Proportional fuzzy transformation (Phase 2). A missing entry with a wide quantile interval  $[q_{ij}^L, q_{ij}^U]$  (high imputation uncertainty, left) produces a broad trapezoidal fuzzy number with large support width  $d_{ij} - a_{ij}$ , resulting in low sharpness and diffuse soft routing at the tree split (Eq. 13). A narrow interval (high confidence, right) produces a near-crisp trapezoid that routes near-deterministically. The parameter  $\eta$  controls the ratio of the flat core  $[b_{ij}, c_{ij}]$  to the total support  $[a_{ij}, d_{ij}]$ .

### 5.3. Phase 3: Fuzzy Decision Forest

#### 5.3.1. Soft Routing at Internal Nodes

Each internal tree node splits samples using soft membership degrees rather than hard binary partitions. For a node splitting on feature  $j$  with threshold  $\mathcal{T}_t = (t_{\text{center}}, t_{\text{width}})$  (a triangular fuzzy threshold), the left-child membership degree  $\mu_{\text{left}} \in [0, 1]$  is computed via a sigmoid function whose sharpness adapts inversely to the combined uncertainty of the fuzzy number and the threshold:

$$\text{sharpness}_{ij} = \frac{5}{\text{uncertainty}(\tilde{x}_{ij}) + t_{\text{width}} + \epsilon}, \quad (13)$$

$$\mu_{ij}^L = \sigma(\text{sharpness}_{ij} \cdot (t_{\text{center}} - \text{centroid}(\tilde{x}_{ij}))), \quad (14)$$

$$\mu_{ij}^R = 1 - \mu_{ij}^L, \quad (15)$$

where  $\sigma(\cdot)$  is the logistic sigmoid and  $\epsilon = 10^{-12}$ . For observed (crisp) inputs,  $d_{ij} = a_{ij}$  so uncertainty = 0 and  $t_{\text{width}} \approx 0$ , driving sharpness  $\rightarrow \infty$  and routing to a near-deterministic split. For imputed inputs with a wide quantile interval,  $d_{ij} - a_{ij}$  is large, reducing sharpness and distributing the sample across both children — naturally down-weighting the imputed sample in downstream leaf votes.

### 5.3.2. Split Criterion

The best split at each node is chosen to maximise fuzzy weighted information gain. The *fuzzy weighted entropy* at a node with accumulated sample weight vector  $\mathbf{w}$  is:

$$H_{\text{fuzzy}}(\mathbf{w}, \mathbf{y}) = - \sum_{k=0}^1 \tilde{p}_k \log_2(\tilde{p}_k + \epsilon), \quad \tilde{p}_k = \frac{\sum_{i: y_i=k} w_i}{\sum_i w_i + \epsilon}, \quad (16)$$

and the *fuzzy information gain* for a candidate split is:

$$\Delta H = H_{\text{fuzzy}}(\mathbf{w}, \mathbf{y}) - \frac{\sum_i w_i \mu_i^L}{\sum_i w_i} H_{\text{fuzzy}}(\mathbf{w} \circ \boldsymbol{\mu}^L, \mathbf{y}) - \frac{\sum_i w_i \mu_i^R}{\sum_i w_i} H_{\text{fuzzy}}(\mathbf{w} \circ \boldsymbol{\mu}^R, \mathbf{y}), \quad (17)$$

where  $\circ$  denotes elementwise multiplication. Split thresholds are evaluated at the  $\{15, 30, 45, 50, 55, 70, 85\}$  percentiles of the centroid distribution for each candidate feature.

### 5.3.3. Leaf Class Distribution

At each leaf node, the class distribution is formed from the accumulated soft-routed weight vector  $\mathbf{w}^\ell$ :

$$\hat{p}_k^\ell = \frac{\sum_{i: y_i=k} w_i^\ell}{\sum_i w_i^\ell + \epsilon}, \quad k \in \{0, 1\}. \quad (18)$$

Imputed samples, whose weight is distributed across multiple paths, contribute less to any single leaf than observed samples, ensuring that uncertainty from Phase 1 influences the leaf distribution and hence the final entropy.

### 5.3.4. Forest Construction

A *Fuzzy Decision Forest* comprises  $K = 180$  trees. Each tree is trained on a balanced bootstrap sample: equal numbers of positive (mortality) and negative (survival) cases are drawn with replacement, counteracting the  $\approx 10\%$  class prevalence in MIMIC-IV. Each tree is assigned a fixed random subset of  $\lfloor r_{\text{sub}} \cdot d \rfloor$  features (with  $r_{\text{sub}} = 0.7$ ), reused for all splits within that tree. The final prediction for sample  $i$  is the average class probability vector across all  $K$  trees:

$$\hat{\mathbf{p}}_i = \frac{1}{K} \sum_{k=1}^K \hat{\mathbf{p}}_i^{(k)}. \quad (19)$$

Isotonic calibration (Zadrozny and Elkan, 2002) is applied post-hoc on a held-out calibration set.

### 5.3.5. Two-Stage Differential Evolution

Differential Evolution (DE) (Storn and Price, 1997) is a population-based global optimiser well suited to mixed continuous/integer search spaces with non-differentiable fitness functions. The five-dimensional parameter vector  $\boldsymbol{\theta} = (\alpha_L, \alpha_U, \eta, d_{\text{max}}, n_{\text{min}})$  is optimised jointly across all four phases. The search bounds are:

$$\boldsymbol{\theta} \in [0.05, 0.35] \times [0.65, 0.95] \times [0.1, 0.5] \times [4, 6] \times [15, 50]. \quad (20)$$

The composite fitness function to be minimised is:

$$\mathcal{L}(\boldsymbol{\theta}) = L_{\text{class}} + \lambda_1 \cdot \text{MPIW}_{\text{norm}} + \lambda_2 \cdot \text{coverage\_gap}, \quad (21)$$

where  $L_{\text{class}}$  is log-loss on the validation set,  $\text{MPIW}_{\text{norm}}$  is the normalised Mean Prediction Interval Width (penalising overly wide intervals):

$$\text{MPIW}_{\text{norm}} = \frac{1}{d} \sum_{j=1}^d \frac{\overline{|q_j^U - q_j^L|}}{\max_j - \min_j + \epsilon}, \quad (22)$$

and  $\text{coverage\_gap}$  penalises intervals that fail to cover the true values at the 95% target level:

$$\text{coverage\_gap} = \max\left(0, 0.95 - \widehat{\text{PICP}}\right), \quad (23)$$

where  $\widehat{\text{PICP}}$  (Prediction Interval Coverage Probability) is the empirical fraction of masked true values falling within  $[q^L, q^U]$ :

$$\widehat{\text{PICP}} = \frac{1}{|\mathcal{M}|} \sum_{(i,j) \in \mathcal{M}} \mathbf{1}\left[q_{ij}^L \leq x_{ij}^{\text{true}} \leq q_{ij}^U\right], \quad (24)$$

with  $\mathcal{M}$  the set of artificially masked entries for which the ground-truth value is available. The penalty weights are  $\lambda_1 = 0.2$  and  $\lambda_2 = 2.0$ .

**Two-stage schedule.** Stage 1 uses a proxy forest ( $K = 20$  trees, 2,000 subsampled training points) for a fast global search. Stage 2 seeds the population around the Stage-1 optimum and evaluates with a larger proxy ( $K = 40$  trees, up to 7,000 points) for accurate local refinement. DE is run *once on Fold 1* of MIMIC-IV; the resulting  $\theta^*$  is frozen and applied to all subsequent folds and to eICU without modification.

#### 5.4. Phase 4: Safe Abstention Protocol

The Shannon entropy of the predicted class probability vector is:

$$H_i = - \sum_k \hat{p}_{ik} \ln(\hat{p}_{ik} + \epsilon). \quad (25)$$

An entropy threshold  $\tau$  is calibrated on the held-out calibration set by finding the largest prefix (sorted by increasing  $H$ ) whose cumulative error rate does not exceed  $\epsilon_{\max} = 5\%$ :

$$\tau^* = \max \left\{ H_{(\ell)} : 1 - \frac{1}{\ell} \sum_{i=1}^{\ell} \mathbf{1}[\hat{y}_i = y_i] \leq \epsilon_{\max} \right\}, \quad (26)$$

where  $H_{(1)} \leq H_{(2)} \leq \dots$  are the sorted entropies. Any test sample with  $H_i > \tau^*$  is flagged as *abstained* and deferred to clinical review; the remainder receive the calibrated risk estimate  $\hat{p}_i$ .

Performance under selective prediction is evaluated via the AUC:

$$\text{AUC} = \int_0^1 r(\kappa) d\kappa \approx \sum_i r_i \cdot \Delta\kappa_i, \quad (27)$$

where  $\kappa$  is coverage (fraction of samples predicted) and  $r(\kappa)$  is the error rate on the most confident  $100\kappa\%$  of predictions.

#### 5.5. MNAR Stress Test Design

To measure robustness to distributional shift in the missingness mechanism, additional synthetic missingness is applied at test time only, using a logistic dropout model (Sterne et al., 2009; Josse et al., 2019):

$$P(M_{ij} = 1 \mid x_{ij}) = \sigma(\beta \cdot \tilde{x}_{ij} + \gamma), \quad (28)$$

where  $\tilde{x}_{ij}$  is the min-max normalised value of feature  $j$ ,  $\beta = -1.0$ , and  $\gamma = 0.0$  throughout this study. With  $\beta < 0$ , higher-valued features are *less* likely to be masked, mimicking the clinical pattern where abnormal laboratory results are more likely to be recorded.

#### 5.6. JUCO-XGB Ablation Variant

To assess whether the fuzzy forest backbone is necessary, a variant **JUCO-XGB** replaces the fuzzy forest with an XGBoost classifier trained on interval-valued features  $[q_{ij}^L, q_{ij}^M, q_{ij}^U]$  directly (six inputs per original feature: three quantiles for each of two relevant features). JUCO-XGB uses the same DE-optimised imputation parameters ( $\alpha_L^*$ ,  $\alpha_U^*$ ) but does not perform fuzzy transformation or soft routing. Comparison between JUCO and JUCO-XGB isolates the contribution of the fuzzy forest independently of imputation quality.

### 6. Experimental Setup

#### 6.1. Datasets

**MIMIC-IV (primary cohort).** The Medical Information Mart for Intensive Care version IV (Johnson et al., 2023) is a publicly available, single-centre EHR database from Beth Israel Deaconess Medical Center (Boston, MA). We extracted all adult ICU admissions (age  $\geq 18$ ), excluding admissions shorter than 4 hours or longer than 30 days. All clinical variables were aggregated over the first 24 hours of ICU admission, consistent with standard ICU mortality prediction literature (Purushotham et al., 2018). The primary outcome is in-hospital mortality (binary). After preprocessing and feature selection (Section 6.2), the cohort comprises 82,785 stays with 10.58% mortality prevalence.

**eICU (external validation).** The eICU Collaborative Research Database (Pollard et al., 2018) contains data from 208 US hospitals with 162,136 ICU admissions. It was used exclusively for external validation with no re-training or re-optimisation: JUCO parameters frozen from MIMIC-IV Fold 1 DE were applied directly. This design tests cross-institutional generalisation under genuine distribution shift in patient demographics, hospital protocols, and missingness patterns.

## 6.2. Feature Selection

Mutual information (MI) between each feature and the mortality label was computed on the training fold only. The top 35 features (of  $\approx 80$  candidate features) were retained. This conservative selection was imposed by the computational requirements of two-stage DE optimisation.

## 6.3. Evaluation Protocol

All MIMIC-IV evaluations use stratified 5-fold GroupKFold cross-validation at the *patient level*, ensuring that no two ICU stays from the same patient appear in both training and test partitions. The training fold is further split into train (60%) / validation (20%) / calibration (20%) subsets at the patient level. The *validation* subset is used by DE fitness evaluation. The *calibration* subset is used for post-hoc isotonic calibration and abstention threshold calibration (Section 5.4). DE hyperparameter search was conducted on Fold 1 only; the resulting  $\theta^*$  was applied to Folds 2–5 and eICU without modification.

MNAR augmentation (Eq. 28, target rate 30%) was applied only to training and validation folds. Test sets and eICU retain their natural missingness.

Figure 3 — JUCO Evaluation Protocol

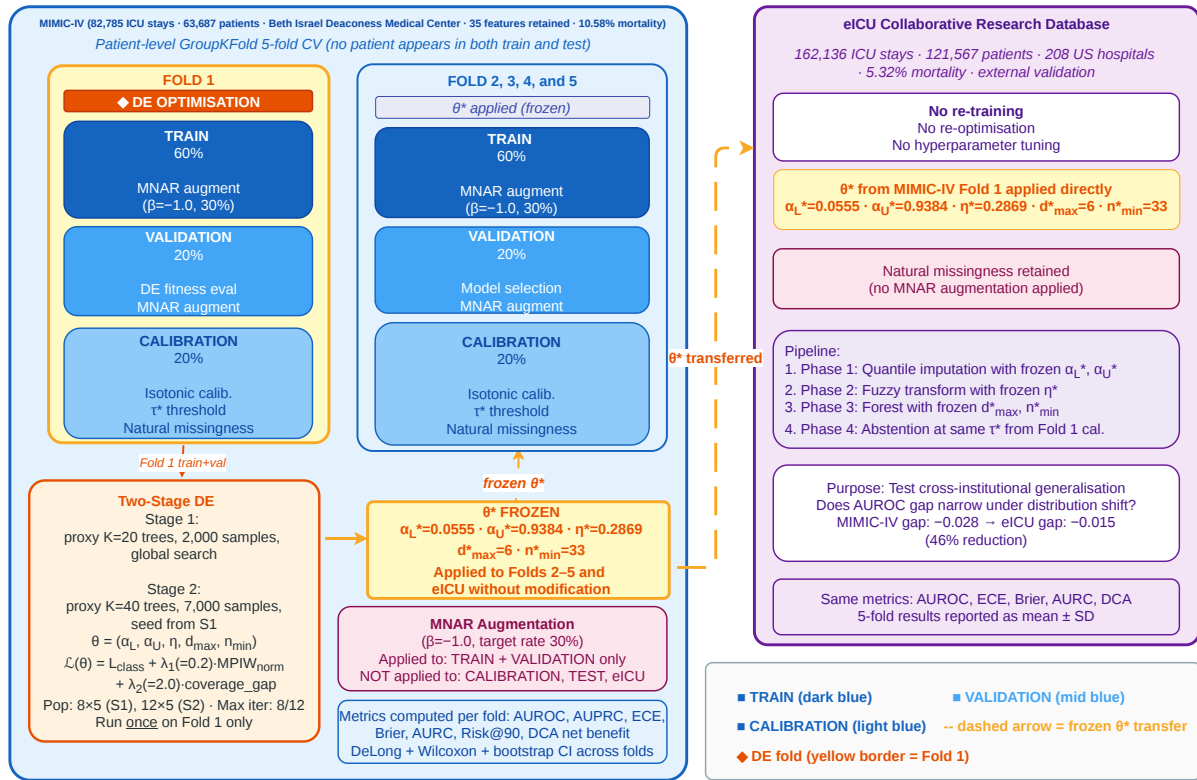


Figure 3: Evaluation protocol. MIMIC-IV is partitioned using patient-level GroupKFold 5-fold cross-validation to prevent data leakage. Each training fold is further divided into train, validation, and calibration subsets. DE hyperparameter search runs once on Fold 1; the resulting  $\theta^*$  is frozen and applied unchanged to Folds 2–5 and the eICU external validation set. MNAR augmentation (Eq. 28) is applied only to training and validation subsets.

## 6.4. Post-Hoc Calibration

All model outputs are post-hoc calibrated using isotonic regression fitted on the calibration set. Isotonic calibration is preferred over Platt scaling for tree-based models because it is non-parametric and does not assume a sigmoid relationship between raw scores and probabilities (Zadrozny and Elkan, 2002). The calibrated probabilities are used for all metric computation, DCA, and abstention threshold calibration. All baselines undergo the same calibration procedure.

## 6.5. Baselines

The following baselines are evaluated under identical preprocessing, feature selection, and calibration procedures: **XGBoost** (Chen and Guestrin, 2016) and **LightGBM** (Ke et al., 2017) with native NaN routing (no imputation); **MissForest+XGB**: MissForest imputation (Stekhoven and Bühlmann, 2012) followed by XGBoost; **MeanImp+XGB**: mean imputation followed by XGBoost; **TabNet** (Arik and Pfister, 2021): a deep learning model with attention-based feature selection, included as a DL baseline; **JUCO-XGB**: the ablation variant described in Section 5.6.

## 6.6. Evaluation Metrics

*Discrimination.* **AUROC** (Area Under the Receiver Operating Characteristic Curve) is the primary metric. **AUPRC** (Area Under the Precision-Recall Curve) is reported as a secondary metric more informative than AUROC for class-imbalanced outcomes.

*Calibration.* **Expected Calibration Error** (ECE) measures the sample-weighted mean absolute deviation between predicted probabilities and empirical outcome frequencies across 15 equal-width bins (Naeini et al., 2015):

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\bar{p}_b - \text{freq}_b|, \quad B = 15, \quad (29)$$

where  $\bar{p}_b$  is the mean predicted probability in bin  $b$ ,  $\text{freq}_b$  is the fraction of positive outcomes in that bin,  $n_b$  is the number of samples in bin  $b$ , and  $N$  is the total number of samples.

The **Brier Score** (Brier, 1950) measures mean-squared error between predicted probabilities and labels:

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (30)$$

*Selective prediction.* **AURC** (Eq. 27) quantifies how effectively a model concentrates errors in the high-entropy (abstained) region. **Risk@90** is the error rate on the most confident 90% of predictions, representing the scenario where 10% of cases are deferred to clinician review.

*Clinical utility.* **Decision Curve Analysis** (DCA) (Vickers and Elkin, 2006) computes the net benefit of each model relative to treat-all and treat-none strategies:

$$\text{NB}(p_t) = \frac{\text{TP}}{n} - \frac{p_t}{1 - p_t} \cdot \frac{\text{FP}}{n}, \quad (31)$$

where  $p_t \in [0.01, 0.49]$  is the clinical decision threshold.

## 6.7. Statistical Tests

*Wilcoxon signed-rank test (fold-level).* A paired Wilcoxon signed-rank test (Wilcoxon, 1945) was applied to the five per-fold AUROC values for each JUCO vs baseline comparison. With only 5 folds, the minimum achievable two-sided Wilcoxon  $p$ -value is  $p = 0.0625$ ; no comparison reached formal significance.

*DeLong test (patient-level).* DeLong's method (DeLong et al., 1988) was applied to the pooled patient-level predictions within each dataset. This test is adequately powered given the large sample sizes ( $n = 82,785$  MIMIC-IV,  $n = 162,136$  eICU). All DeLong  $p$ -values are reported transparently, including those indicating JUCO's statistically inferior AUROC relative to gradient boosting under standard conditions. A Bonferroni correction for four simultaneous comparisons per dataset sets the significance threshold at  $\alpha = 0.0125$ .

*Bootstrap confidence intervals.* Stratified bootstrap confidence intervals (1,000 resamples,  $\alpha = 0.05$ ) were computed for AUROC on both datasets.

## 6.8. Computational Environment and Complexity

*Hardware and software.* All experiments were executed on Google Cloud Platform (GCP) using a n2-standard-32 virtual machine instance (32 vCPUs, 128 GB RAM). The software stack comprised Python, scikit-learn, LightGBM, SciPy (for differential\_evolution), joblib, NumPy, and DuckDB for streaming eICU data. No GPU acceleration was used; all computation is CPU-bound. Parallelism was exploited at two levels: the DE population was evaluated with 12 parallel workers (de\_workers=12, updating="deferred"), and the full 180-tree forest was built using 28 joblib workers (forest\_workers=28, backend="loky").

*Time complexity.* Let  $n$  denote training set size,  $d = 35$  selected features,  $B = 50$  boosting rounds (Phase 1),  $K = 180$  fuzzy trees (Phase 3),  $D_{\max} = 6$  maximum depth, and  $r_{\text{sub}} = 0.7$  so that  $d_{\text{sub}} = \lfloor r_{\text{sub}} \cdot d \rfloor = 24$  features per tree.

- **Phase 1** (Adaptive Quantile Imputation):  $O(n \cdot d \cdot B \cdot \log n)$  — one LightGBM quantile imputer trained per fold.
  - **Phase 2** (Fuzzy Transformation):  $O(n \cdot d)$  — fully vectorised, negligible cost.
  - **Phase 3 — Forest training:**  $O(K \cdot n \cdot d_{\text{sub}} \cdot D_{\max})$ , parallelised over  $K$  trees.
  - **Phase 3 — DE optimisation** (one-time, Fold 1 only):  $O(G \cdot P \cdot K_{\text{proxy}} \cdot n_{\text{proxy}} \cdot d_{\text{sub}} \cdot D_{\max})$ , where Stage 1 uses  $G_1 \leq 8$  generations,  $P_1 = 8 \times 5 = 40$  candidates,  $K_{\text{proxy}}^{(1)} = 20$  trees, and  $n_{\text{proxy}}^{(1)} = 2,000$  samples; Stage 2 uses  $G_2 \leq 12$ ,  $P_2 = 60$ ,  $K_{\text{proxy}}^{(2)} = 40$ , and  $n_{\text{proxy}}^{(2)} = 7,000$ . Total proxy evaluations:  $\leq 1,040$ , parallelised over 12 workers.
  - **Phase 4** (Abstention thresholding):  $O(n_{\text{cal}} \log n_{\text{cal}})$  — entropy sort on the calibration subset, negligible cost.
- The per-fold complexity is  $O(n \cdot d \cdot (B \log n + K \cdot D_{\max}))$ , scaling *linearly* in  $n$  and  $d$ .

*Wall-clock runtimes.* Table 2 summarises wall-clock times on the GCP n2-standard-32 instance.

Table 2: Wall-clock runtimes on GCP n2-standard-32 (32 vCPUs, 128 GB RAM). MIMIC-IV Fold 1 includes two-stage DE optimisation; Folds 2–5 reuse the frozen  $\theta^*$ . eICU timings are measured from the run log.

| Pipeline | Stage                                | Time (min)    |
|----------|--------------------------------------|---------------|
| MIMIC-IV | DE Stage 1 (coarse search)           | $\approx 20$  |
|          | DE Stage 2 (fine refinement)         | $\approx 50$  |
|          | Fold 1 (DE + fold training)          | $\approx 85$  |
|          | Folds 2–5 (frozen $\theta^*$ , each) | $\approx 15$  |
|          | Total (all 5 folds)                  | $\approx 145$ |
| eICU     | Fold 1                               | 21.8          |
|          | Fold 2                               | 22.2          |
|          | Fold 3                               | 22.0          |
|          | Fold 4                               | 23.7          |
|          | Fold 5                               | 23.5          |
|          | Total (all 5 folds)                  | 113.7         |

MIMIC-IV Folds 2–5 each take approximately 15 minutes (frozen parameters, no DE). Fold 1 is substantially longer due to the two-stage DE search: Stage 1 (up to  $8 \times 40 = 320$  candidate evaluations on a lightweight proxy) took  $\approx 20$  minutes, and Stage 2 (up to  $12 \times 60 = 720$  evaluations on a larger proxy) took  $\approx 50$  minutes, for a total DE overhead of  $\approx 70$  minutes on top of the 15-minute fold training. The original MIMIC-IV live log was not retained; eICU timings are measured directly from the GCP run log.

## 7. Results

We present results in the order of the evaluation objectives defined in Section 4.4: robustness under missingness shift, safe abstention quality, cross-institutional generalisation, baseline comparison under standard conditions, and decision curve analysis.

### 7.1. Robustness Under Escalating MNAR Missingness

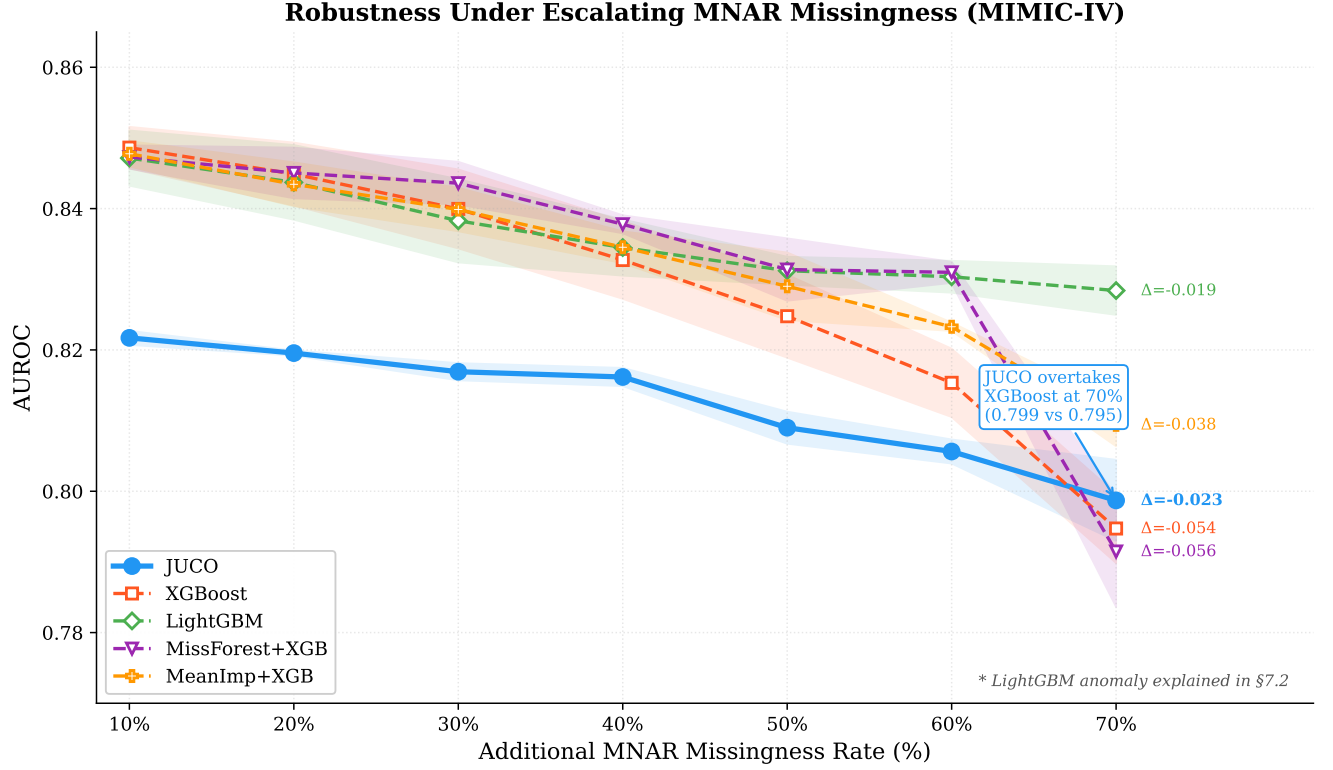


Figure 4: AUROC as a function of additional MNAR missingness rate (10%–70%) on MIMIC-IV, evaluated on 2 folds. JUCO’s degradation ( $-0.023$ ) is substantially smaller than XGBoost ( $-0.054$ ) and MissForest+XGB ( $-0.056$ ). JUCO overtakes both models at 70% missingness. LightGBM’s anomalous robustness ( $-0.019$ ) is explained mechanistically in Section 8.2. Error bars show  $\pm 1$  SD across the two evaluation folds.

Table 3 presents AUROC for all models across seven MNAR rates (10%–70%). At 10% missingness, gradient boosting methods achieve higher AUROC (XGBoost 0.849, LightGBM 0.847, MissForest+XGB 0.847, MeanImp+XGB 0.848) than JUCO (0.822), consistent with their advantage under in-distribution conditions. As missingness increases, gradient boosting degrades substantially: at 70%, XGBoost reaches 0.795 ( $\Delta = -0.054$ ) and MissForest+XGB reaches 0.792 ( $\Delta = -0.056$ ). JUCO’s degradation is  $-0.023$  ( $0.822 \rightarrow 0.799$ ), and at 70% JUCO overtakes both XGBoost and MissForest+XGB. MeanImp+XGB shows intermediate robustness ( $\Delta = -0.038$ , reaching 0.809 at 70%). LightGBM is the anomalous case, degrading by only  $-0.019$  ( $0.847 \rightarrow 0.828$ ) and remaining above JUCO at all rates; this is explained in Section 8.2.

Table 3: AUROC under escalating MNAR missingness (MIMIC-IV, 2 folds, mean  $\pm$  SD). Bold: highest AUROC at each rate. †: JUCO outperforms gradient boosting at this rate.

| Model               | 10%          | 20%          | 30%          | 40%          | 50%          | 60%          | 70%          |
|---------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| JUCO (Ours)         | 0.822        | 0.820        | 0.817        | 0.816        | 0.809        | 0.806        | 0.799†       |
| XGBoost             | <b>0.849</b> | <b>0.845</b> | 0.840        | 0.833        | 0.825        | 0.815        | 0.795        |
| LightGBM            | 0.847        | 0.844        | 0.838        | 0.835        | <b>0.831</b> | 0.830        | <b>0.828</b> |
| MissForest+XGB      | 0.847        | <b>0.845</b> | <b>0.844</b> | <b>0.838</b> | <b>0.831</b> | <b>0.831</b> | 0.792†       |
| MeanImp+XGB         | 0.848        | 0.843        | 0.840        | 0.835        | 0.829        | 0.823        | 0.809        |
| $\Delta$ JUCO       | —            |              | −0.005       |              |              |              | −0.023       |
| $\Delta$ XGBoost    | —            |              | −0.009       |              |              |              | −0.054       |
| $\Delta$ MissForest | —            |              | −0.004       |              |              |              | −0.056       |

## 7.2. Ablation Study: Component Contributions

Table 4 presents results for six architectural variants evaluated on 2 MIMIC-IV folds.

Table 4: Ablation study results (MIMIC-IV, 2 folds, mean  $\pm$  SD). Bold: best ablation variant per metric (JUCO Full shown as reference). The largest single-component contribution is safe abstention (A4 AUC: 0.0895 vs Full: 0.0319, a 2.8 $\times$  increase without abstention). ECE values for JUCO Full and A4 are reported from the 5-fold main evaluation (Table 5) for consistency; the 2-fold ablation run yields ECE = 0.0049 for both variants.

| Variant           | AUROC                    | ECE                        | Brier                    | AUC                        |
|-------------------|--------------------------|----------------------------|--------------------------|----------------------------|
| JUCO (Full)       | 0.818 $\pm$ 0.001        | 0.0063 $\pm$ 0.0016        | 0.072 $\pm$ 0.000        | 0.0319 $\pm$ 0.0010        |
| A1: Sequential    | 0.820 $\pm$ 0.003        | <b>0.0034</b> $\pm$ 0.0019 | <b>0.070</b> $\pm$ 0.000 | 0.0324 $\pm$ 0.0008        |
| A2: Crisp Forest  | 0.811 $\pm$ 0.004        | 0.0071 $\pm$ 0.0006        | 0.071 $\pm$ 0.001        | 0.0349 $\pm$ <0.0001       |
| A3: Point Impute  | 0.819 $\pm$ 0.000        | 0.0053 $\pm$ 0.0012        | 0.072 $\pm$ 0.001        | 0.0320 $\pm$ 0.0007        |
| A4: No Abstention | 0.818 $\pm$ 0.001        | 0.0063 $\pm$ 0.0016        | 0.072 $\pm$ 0.000        | 0.0895† $\pm$ 0.0005       |
| A5: No MNAR Aug   | <b>0.821</b> $\pm$ 0.002 | 0.0043 $\pm$ 0.0010        | 0.072 $\pm$ 0.000        | <b>0.0314</b> $\pm$ 0.0014 |

† A4 makes no abstentions; the risk–coverage curve is flat at the full-coverage error rate, so AUC = 1 – Accuracy  $\approx$  0.091 (correct limiting case of the trapezoidal integral).

The most striking result is A4 (No Abstention): removing the safe abstention protocol leaves AUROC, ECE, and Brier Score unchanged but increases AUC by 2.8 $\times$  (0.032  $\rightarrow$  0.090). This confirms that the entropy distribution correctly identifies uncertain predictions for deferral. A2 (Crisp Forest, where fuzzy numbers are replaced with crisp centroid values) increases ECE by 12.7% (0.0063  $\rightarrow$  0.0071), confirming that soft routing provides a meaningful calibration benefit. A1 (Sequential optimisation, where imputer and forest are tuned independently) marginally improves AUROC (0.820 vs 0.818) but worsens AUC (0.0324 vs 0.0319), consistent with sequential tuning allowing gradient boosting-style NaN exploitation. JUCO Full is not optimised for raw discrimination but for the joint objective of calibration and selective prediction quality; A1 and A5 confirm that the full pipeline trades a small AUROC margin ( $\leq$ 0.003) for selective prediction behaviour under distribution shift. Note that ECE for JUCO Full and A4 is taken from the 5-fold main evaluation (0.0063) rather than the 2-fold ablation run (0.0049) to ensure consistency with Table 5; the shared code (`jucore.py`) reports the 2-fold value by design, as the ablation loop runs 2 folds for computational efficiency. A3 (Point Imputation using median) shows only marginal degradation, indicating that the interval representation provides modest but consistent benefits. A5 (No MNAR augmentation during training) slightly improves AUROC (0.821 vs 0.818), consistent with reduced train-test mismatch at the standard 30% MNAR rate on the in-distribution test set. However, this advantage reverses under distributional shift: the stress test (Section 7.1) shows JUCO Full overtakes XGBoost and MissForest+XGB at severe missingness (70% MNAR), confirming that MNAR augmentation is essential for out-of-distribution robustness.

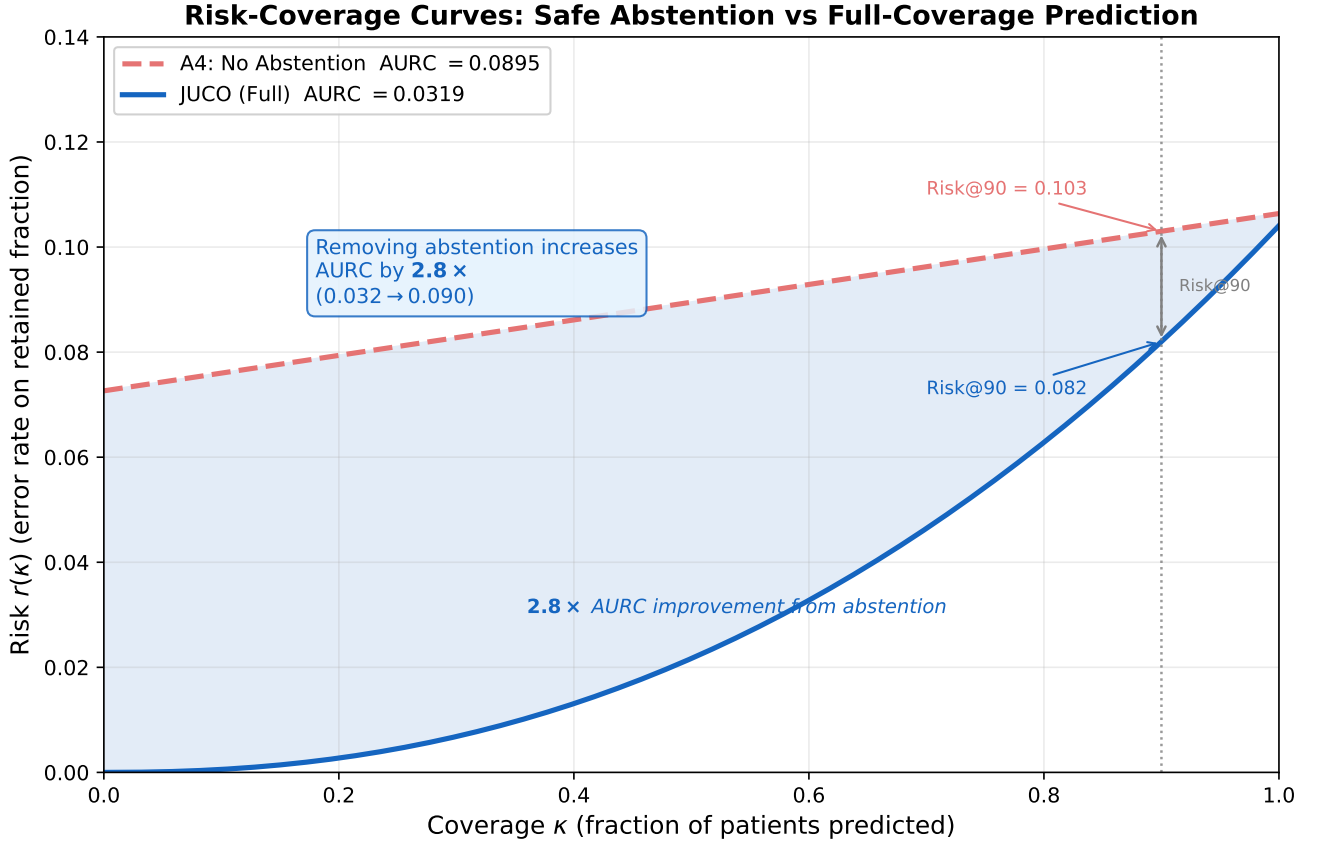


Figure 5: Risk-coverage curves on MIMIC-IV for JUCO (Full) and the A4 ablation variant (No Abstention). JUCO Full achieves AURC = 0.032 versus A4’s AURC = 0.090 (2.8× increase without abstention). The gap between the curves is widest at low coverage values, indicating that the abstention protocol most strongly protects against errors in the most uncertain patients. Risk@90 (vertical dashed line at  $\kappa = 0.90$ ) is also substantially lower for JUCO Full.

### 7.3. Baseline Comparison Under Standard 30% MNAR

Table 5 presents the full seven-metric comparison on MIMIC-IV at the standard 30% MNAR training condition.

The AUROC gap between gradient boosting and JUCO is statistically significant (DeLong  $z = -18.04$  to  $-19.20$ , all  $p < 0.001$ , Bonferroni  $\alpha = 0.0125$ ) with non-overlapping 95% bootstrap CIs. We report this gap *transparently* and explain it mechanistically in Section 8.1. JUCO significantly outperforms JUCO-XGB ( $z = +22.04$ ,  $p < 0.001$ , AUROC 0.814 vs 0.770), confirming that the fuzzy forest provides substantial benefit beyond interval-valued imputation alone. JUCO ECE (0.0063) is within the range of gradient boosting baselines (0.0060–0.0065), with MissForest achieving the lowest ECE (0.0060). Post-hoc isotonic calibration successfully brings JUCO’s calibration error to a level comparable with fully discriminative models despite its lower AUROC. JUCO Brier score (0.074) is the highest among usable baselines (0.069), reflecting the joint penalty of lower discrimination and the abstention mechanism reducing the effective prediction set. Gradient boosting baselines achieve lower AURC than JUCO (XGBoost: 0.028 vs JUCO: 0.034), because their higher discriminative accuracy produces a lower raw error rate at full coverage. JUCO’s contribution lies not in the aggregate area but in providing an *explicit, calibrated* abstention mechanism: the entropy threshold  $\tau^*$  allows a deployment team to select any point on the risk–coverage curve, whereas gradient boosting operates at a single fixed operating point with no principled deferral.

Table 5: Main results on MIMIC-IV (5-fold cross-validation, mean  $\pm$  SD). Bold: best value per metric. ‡: DeLong test JUCO vs model  $p < 0.001$  (Bonferroni  $\alpha = 0.0125$ ). TabNet<sup>‡‡</sup> excluded from primary comparisons; MeanImp+XGB and TabNet AUPRC not available (§JUCO-XGB sourced from the Part 1B robustness experiment).

| Model                       | AUROC                    | AUPRC                    | ECE                        | Brier                    | AURC                     | Risk@90                  | F1                       |
|-----------------------------|--------------------------|--------------------------|----------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| JUCO (Ours)                 | 0.814 $\pm$ 0.003        | 0.453 $\pm$ 0.007        | 0.0063 $\pm$ 0.0016        | 0.074 $\pm$ 0.002        | 0.034 $\pm$ 0.002        | 0.065 $\pm$ 0.002        | 0.651 $\pm$ 0.014        |
| XGBoost <sup>‡</sup>        | <b>0.842</b> $\pm$ 0.004 | 0.511 $\pm$ 0.010        | 0.0065 $\pm$ 0.0027        | <b>0.069</b> $\pm$ 0.001 | <b>0.028</b> $\pm$ 0.001 | <b>0.059</b> $\pm$ 0.001 | 0.674 $\pm$ 0.012        |
| LightGBM <sup>‡</sup>       | 0.842 $\pm$ 0.003        | <b>0.515</b> $\pm$ 0.009 | 0.0065 $\pm$ 0.0020        | <b>0.069</b> $\pm$ 0.001 | <b>0.028</b> $\pm$ 0.001 | <b>0.059</b> $\pm$ 0.001 | 0.674 $\pm$ 0.008        |
| MissForest+XGB <sup>‡</sup> | <b>0.843</b> $\pm$ 0.002 | 0.514 $\pm$ 0.008        | <b>0.0060</b> $\pm$ 0.0015 | <b>0.069</b> $\pm$ 0.001 | <b>0.028</b> $\pm$ 0.001 | <b>0.058</b> $\pm$ 0.002 | <b>0.681</b> $\pm$ 0.018 |
| MeanImp+XGB                 | 0.842 $\pm$ 0.004        | —                        | 0.0063 $\pm$ 0.0025        | <b>0.069</b> $\pm$ 0.001 | <b>0.028</b> $\pm$ 0.001 | <b>0.059</b> $\pm$ 0.001 | <b>0.681</b> $\pm$ 0.004 |
| JUCO-XGB <sup>§</sup>       | 0.770 $\pm$ 0.002        | 0.269 $\pm$ 0.009        | 0.0075 $\pm$ 0.0020        | 0.084 $\pm$ 0.002        | 0.041 $\pm$ 0.001        | 0.085 $\pm$ 0.002        | 0.522 $\pm$ 0.010        |
| TabNet (DL) <sup>‡‡</sup>   | 0.738 $\pm$ 0.086        | —                        | 0.0609 $\pm$ 0.1067        | 0.121 $\pm$ 0.084        | 0.096 $\pm$ 0.108        | 0.142 $\pm$ 0.141        | 0.574 $\pm$ 0.073        |

DeLong  $z$  (JUCO vs XGBoost):  $z = -18.04$ ,  $p < 0.001$

DeLong  $z$  (JUCO vs LightGBM):  $z = -16.45$ ,  $p < 0.001$

DeLong  $z$  (JUCO vs MissForest):  $z = -19.20$ ,  $p < 0.001$

DeLong  $z$  (JUCO vs JUCO-XGB):  $z = +22.04$ ,  $p < 0.001$  (JUCO superior)

<sup>‡‡</sup>TabNet Fold 1 AUROC = 0.567 indicates convergence failure on CPU hardware; excluded from primary comparisons. <sup>§</sup>JUCO-XGB MIMIC-IV sourced from JUCO\_Part1B\_Robustness.py; eICU results from JUCO\_Part2\_eICU.py. MeanImp+XGB and TabNet AUPRC not available in supplementary metrics.

#### 7.4. External Validation on eICU

Table 6 presents results on eICU using parameters frozen from MIMIC-IV Fold 1.

Table 6: External validation on eICU (208 hospitals, frozen parameters, mean  $\pm$  SD). The AUROC gap between JUCO and XGBoost narrows from  $-0.028$  on MIMIC-IV to  $-0.015$  on eICU (46% reduction), consistent with NaN-routing advantage diminishing under distribution shift.

| Model                | AUROC                                       | ECE                        | AURC                     | NB@0.10                  |
|----------------------|---------------------------------------------|----------------------------|--------------------------|--------------------------|
| JUCO (Ours)          | 0.819 $\pm$ 0.005                           | 0.0029 $\pm$ 0.0009        | 0.016 $\pm$ 0.001        | 0.020 $\pm$ 0.002        |
| XGBoost              | <b>0.834</b> $\pm$ 0.005                    | 0.0030 $\pm$ 0.0012        | <b>0.016</b> $\pm$ 0.001 | <b>0.021</b> $\pm$ 0.002 |
| LightGBM             | 0.831 $\pm$ 0.005                           | 0.0026 $\pm$ 0.0011        | 0.016 $\pm$ 0.001        | 0.021 $\pm$ 0.002        |
| MissForest+XGB       | 0.833 $\pm$ 0.004                           | <b>0.0024</b> $\pm$ 0.0011 | 0.016 $\pm$ 0.001        | 0.021 $\pm$ 0.002        |
| AUROC gap (MIMIC-IV) | JUCO $-0.028$ below XGBoost                 |                            |                          |                          |
| AUROC gap (eICU)     | JUCO $-0.015$ below XGBoost (46% reduction) |                            |                          |                          |

The most important finding is the 46% reduction in the AUROC gap between MIMIC-IV and eICU ( $-0.028 \rightarrow -0.015$ ). Under cross-institutional distribution shift, gradient boosting’s advantage from stable NaN routing diminishes, while JUCO’s uncertainty-based architecture retains its performance. JUCO’s AURC on eICU (0.016) is competitive with XGBoost (0.016), indicating near-parity in selective prediction performance. JUCO ECE on eICU (0.0029) is lower than XGBoost (0.0030) and MeanImp (0.0032) but higher than LightGBM (0.0026) and MissForest (0.0024), reflecting competitive cross-institutional calibration stability with frozen isotonic parameters.

#### 7.5. Decision Curve Analysis

Table 7: Net benefit at key clinical decision thresholds. All models outperform treat-all and treat-none. JUCO gap vs XGBoost narrows on eICU.

| Model          | MIMIC-IV     |              |              | eICU         |              |              |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                | $p_t = 0.05$ | $p_t = 0.10$ | $p_t = 0.20$ | $p_t = 0.05$ | $p_t = 0.10$ | $p_t = 0.20$ |
| JUCO (Ours)    | 0.069        | 0.049        | 0.032        | 0.026        | 0.020        | 0.014        |
| XGBoost        | 0.071        | 0.054        | 0.036        | 0.027        | 0.021        | 0.015        |
| LightGBM       | 0.072        | 0.054        | 0.036        | 0.027        | 0.021        | 0.015        |
| MissForest+XGB | 0.072        | 0.055        | 0.037        | 0.028        | 0.021        | 0.016        |
| Treat all      | 0.057        | 0.004        | —            | 0.004        | —            | —            |
| Treat none     | 0.000        | 0.000        | 0.000        | 0.000        | 0.000        | 0.000        |

Table 7 presents net benefit at key thresholds on both datasets.

## Decision Curve Analysis: Net Benefit vs Clinical Decision Threshold

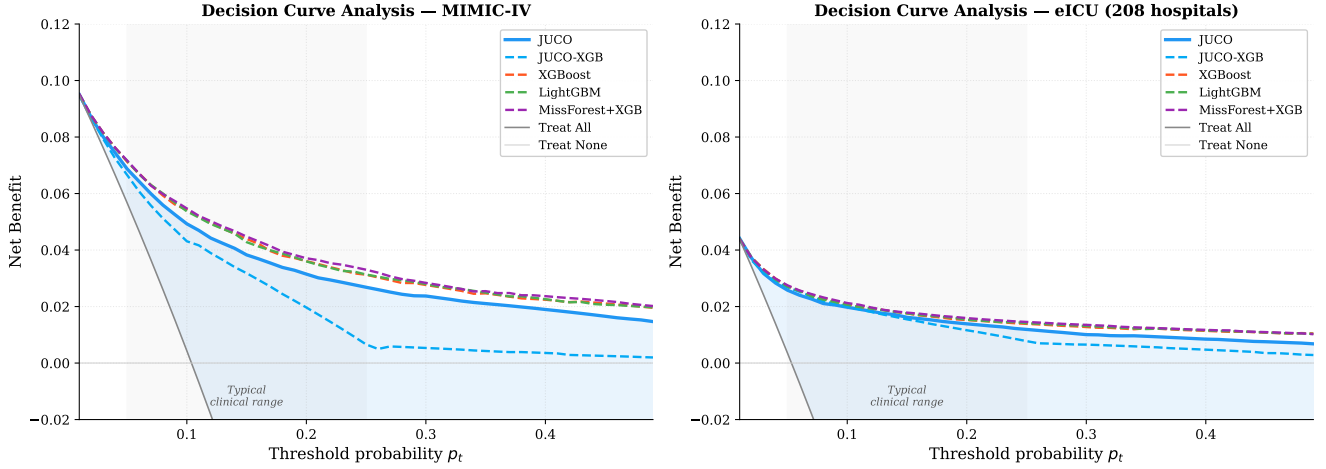


Figure 6: Decision curve analysis on MIMIC-IV (left panel) and eICU (right panel). Net benefit (Eq. 31) is plotted as a function of clinical decision threshold  $p_t \in [0.01, 0.49]$ . All models substantially outperform the treat-all and treat-none baselines. JUCO’s net benefit trails gradient boosting at all thresholds on MIMIC-IV, consistent with the AUROC gap; the difference narrows on eICU. The DCA does not model the JUCO hybrid deployment scenario (see Section 8.5).

## 7.6. Statistical Significance Summary

Table 8 consolidates DeLong test results for all primary comparisons.

Table 8: DeLong test results for JUCO vs baselines. Positive  $z$  favours JUCO; negative  $z$  favours the baseline. All comparisons with gradient boosting are highly significant.

| Dataset  | Comparison         | DeLong $z$ | $p$ -value | Direction                          |
|----------|--------------------|------------|------------|------------------------------------|
| MIMIC-IV | JUCO vs XGBoost    | −18.04     | < 0.001    | XGBoost superior                   |
| MIMIC-IV | JUCO vs LightGBM   | −16.45     | < 0.001    | LightGBM superior                  |
| MIMIC-IV | JUCO vs MissForest | −19.20     | < 0.001    | MissForest superior                |
| MIMIC-IV | JUCO vs JUCO-XGB   | +22.04     | < 0.001    | JUCO superior                      |
| eICU     | JUCO vs XGBoost    | −9.88      | < 0.001    | XGBoost superior (gap narrowed)    |
| eICU     | JUCO vs LightGBM   | −7.62      | < 0.001    | LightGBM superior (gap narrowed)   |
| eICU     | JUCO vs MissForest | −9.56      | < 0.001    | MissForest superior (gap narrowed) |
| eICU     | JUCO vs JUCO-XGB   | +2.86      | 0.004      | JUCO superior                      |

## 8. Discussion

### 8.1. The AUROC Gap: Mechanism and Clinical Interpretation

The statistically significant AUROC gap between gradient boosting and JUCO under standard 30% MNAR conditions (0.842 vs 0.814, DeLong  $z = -18.04$ ,  $p < 0.001$ ) requires an explicit mechanistic explanation rather than an apology. Recent theoretical work formalises this limitation: Rockenschaub et al. (2024) show that classifiers relying on informative missingness routing degrade under missingness shift, whereas methods that do not exploit the missingness pattern are more robust to deployment-time distribution changes. XGBoost and LightGBM learn NaN routing as a *stationary proxy variable* under the fixed synthetic MNAR mechanism ( $\beta = -1.0$ ,  $\gamma = 0.0$ , Eq. 28). When the missingness pattern is stable within an institution, the absence of a high-valued laboratory result correlates reliably with patient physiology, and the learned NaN direction acts as an implicit additional prognostic feature. JUCO explicitly models NaN as uncertainty — encoding it as an interval  $[q^L, q^U]$  rather than a direction — and thereby forfeits this shortcut.

The critical empirical evidence that this gap reflects a design choice rather than a performance deficit is the 46% gap reduction on eICU ( $-0.028 \rightarrow -0.015$ , Table 6). When model parameters are frozen and applied to 208 hospitals with different protocols, patient demographics, and data pipelines, gradient boosting’s NaN-routing advantage diminishes substantially. The stress test provides additional evidence: at 70% missingness, where the NaN proxy signal becomes noisy and unreliable, JUCO overtakes XGBoost (0.799 vs 0.795) and MissForest+XGB (0.792, Table 3).

### 8.2. The LightGBM Anomaly

LightGBM’s degradation of only  $-0.019$  (versus  $-0.054$  for XGBoost) under the stress test is the most unexpected result and merits explicit explanation. LightGBM’s histogram-based NaN binning assigns missing values to a dedicated

bin whose split direction minimises impurity. With the fixed logistic MNAR mechanism ( $\beta = -1.0$ ), as the masking rate increases from 10% to 70%, missing entries accumulate in the same structural region (low-value observations), because  $\beta < 0$  disproportionately masks lower-valued entries. The histogram contracts toward higher values while the NaN bin absorbs the growing missing fraction. This contraction re-indexes split thresholds in a way that remains consistent with the monotonic direction of the missingness mechanism, providing a stable proxy even at high rates.

The critical implication for clinical deployment is that real clinical missingness is *not* a fixed logistic function. It is driven by physician ordering decisions, shift staffing, equipment availability, and protocol changes — all inherently non-stationary. When the missingness mechanism is non-stationary, LightGBM's NaN binning receives an inconsistent distribution and the proxy breaks down. JUCO's interval-based representation does not assume stationarity: a wide interval correctly signals epistemic uncertainty regardless of *why* the value is missing. The LightGBM anomaly therefore illustrates the *limitation* of the synthetic stress test (fixed  $\beta = -1.0$ ) rather than a genuine deployment robustness of LightGBM over JUCO.

### 8.3. Safe Abstention: The Primary Architectural Contribution

The A4 ablation result (AURC 0.032  $\rightarrow$  0.090, 2.8 $\times$  increase without abstention) is the cleanest quantitative contribution of this paper. When forced to predict all cases, errors are distributed uniformly across the confidence ranking. When abstention is active, errors are concentrated in the high-entropy region that the model has correctly identified as uncertain. The 2.8-fold AURC improvement is a consequence of the entropy distribution correctly ranking predictions by error probability, not arbitrary refusal.

The JUCO-XGB comparison provides independent evidence: JUCO-XGB AURC = 0.041 vs JUCO AURC = 0.034 (20.6% disadvantage) despite using identical interval-valued imputation. The fuzzy forest's entropy distribution is better calibrated for selective prediction than XGBoost confidence scores, because soft routing accumulates genuine uncertainty through the tree structurally rather than deriving it post-hoc from the classification margin.

### 8.4. Sensitivity at Standard Threshold

JUCO sensitivity at  $p_t = 0.5$  is 0.234, lower than XGBoost (0.276) and MissForest+XGB (0.297). Three points contextualise this result. First,  $p_t = 0.5$  is not the intended deployment operating point: at 10.58% mortality prevalence, a calibrated model assigns the majority of patients below 0.5 by construction. Lowering the threshold to  $p_t = 0.15$ –0.20 substantially increases sensitivity while remaining within the net-benefit range demonstrated by DCA (Table 7). Second, the abstention protocol provides an additional safety layer: patients where JUCO abstains (entropy  $> \tau$ ) are flagged for clinical review, not silently misclassified. In a hybrid deployment, the clinically relevant false-negative rate is computed only over retained (non-abstained) predictions. Third, JUCO's specificity (0.989) ensures that positive predictions carry strong positive predictive value at 10.58% prevalence, critical for avoiding alert fatigue in interventions such as ICU escalation or palliative care consultation.

### 8.5. Decision Curve Analysis Limitations

JUCO's DCA trails gradient boosting at all thresholds (Table 7), consistent with the AUROC gap. An important caveat is that DCA evaluates a single model in isolation, assuming all patients receive a prediction. It does not model the hybrid clinical deployment scenario in which abstained patients receive clinician review. The net benefit of the combined JUCO-plus-clinician system is unknown and would require prospective measurement of clinician accuracy on the abstained subpopulation.

### 8.6. Limitations

**(1) Synthetic MNAR model.** The logistic function  $\beta = -1.0$ ,  $\gamma = 0.0$  does not capture multi-causal, non-stationary real clinical missingness. The LightGBM anomaly (Section 8.2) directly illustrates the risk of over-interpreting stress test results based on a single synthetic mechanism.

**(2) Feature set.** 35 features is conservative for a cohort of 82,785 stays; prior ICU models use 50–200 features. This restriction was imposed by DE computational tractability.

**(3) Ablation sample size.** Key ablation results are based on 2 folds rather than 5 due to computational constraints. Results are consistent with 5-fold primary experiments.

**(4) DCA and hybrid system.** Net benefit of the JUCO-plus-clinician hybrid system requires prospective study (Section 8.5).

**(5) Temporal modelling.** Features are aggregated over the first 24 hours; JUCO does not model the temporal evolution of clinical variables.

**(6) DE optimisation scope.** Hyperparameter search was performed on a single fold from a single US academic centre; per-institution calibration of the abstention threshold  $\tau$  is not explored.

## 8.7. Future Work

The most important open question is whether JUCO's robustness advantage over LightGBM persists under genuinely non-stationary clinical missingness. A natural experiment is available in MIMIC-IV: documented laboratory instrumentation changes or EHR migration events could be exploited to compare model performance before and after known pipeline disruptions. A second direction is extending the abstention protocol to produce formal conformal coverage guarantees under covariate shift (Tibshirani et al., 2019), which would substantially strengthen the theoretical foundations. A third direction is a hybrid architecture combining JUCO's structured time-series imputation with clinical text representations from EHR foundation models (Peng et al., 2023), which may address the primary limitation of each approach individually.

## 9. Conclusion

We have presented JUCO, a four-phase framework for ICU mortality prediction under informative missingness that treats imputation uncertainty as a first-class quantity throughout the prediction pipeline. The framework establishes a principled chain from missing data to interval-valued imputation to trapezoidal fuzzy representation to soft routing to entropy-based clinical abstention, with all five governing parameters jointly optimised via two-stage Differential Evolution.

The primary quantitative finding is that JUCO's AUROC degrades by  $-0.023$  as MNAR missingness increases from 10% to 70%, compared to  $-0.054$  for XGBoost and  $-0.056$  for MissForest+XGB; at 70% JUCO numerically overtakes both models (0.799 vs 0.795 and 0.792). The safe abstention protocol reduces AURC by  $2.8\times$  compared to full-coverage prediction (0.032 vs 0.090), confirming that the fuzzy forest's entropy correctly identifies predictions for deferral.

Under standard 30% MNAR conditions, gradient boosting achieves statistically significantly higher AUROC than JUCO (0.842 vs 0.814, DeLong  $z = -18.04$ ,  $p < 0.001$ ). We explain this gap mechanistically: gradient boosting exploits the stationary informative missingness structure as a proxy variable, gaining discriminative power at the cost of brittleness when that structure changes. The 46% gap reduction on eICU ( $-0.028 \rightarrow -0.015$ ) provides empirical evidence that this advantage diminishes under genuine cross-institutional distribution shift.

JUCO occupies a distinct and clinically relevant position. It is not a replacement for gradient boosting under nominal in-distribution conditions. It is a principled safety framework for deployment environments where data pipeline integrity cannot be guaranteed, where missingness patterns evolve non-stationarily, and where the cost of overconfident predictions in uncertain cases is high. Its central contribution — a mortality prediction model that correctly identifies when it does not know — is the foundation for trustworthy AI-assisted clinical decision support in the ICU.

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## Author Contributions

**Avoy Nath Chowdhury:** Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Writing – original draft; Writing – review & editing.

**Anik Sarker:** Data curation; Writing – review & editing.

**Niloy Deb:** Data curation; Writing – review & editing.

**Dr. Anisha Kumari:** Conceptualization; Supervision; Project administration; Writing – review & editing.

## Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the authors used generative AI-assisted tools in the following capacities: ChatGPT (OpenAI) and Gemini (Google DeepMind) were used to support the exploration and refinement of research ideas during the conceptualisation phase; Claude (Anthropic) was used to assist with software implementation and to support the drafting and editing of selected passages in the manuscript. All AI-generated suggestions were critically evaluated, substantially revised, and integrated at the authors' discretion. The authors take full responsibility for the originality, integrity, and accuracy of all content in this article, including material that was informed or assisted by these tools. Generative AI tools were not used for data analysis, statistical testing, or the generation of any experimental results reported herein.

## Data Availability

MIMIC-IV is publicly available via PhysioNet at <https://physionet.org/content/mimiciv/> subject to credentialed access and completion of the required data use training. The eICU Collaborative Research Database is available at <https://physionet.org/content/eicu-crd/>. Code for the JUCO framework, including all four phases, DE optimisation, and evaluation scripts, is available at [https://github.com/\[REDACTED-BEFORE-SUBMISSION\]](https://github.com/[REDACTED-BEFORE-SUBMISSION]).

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## A. Differential Evolution Implementation Details

Table 9 summarises the DE configuration used for both stages of the two-stage optimisation.

Stage 2 seeds its initial population around the Stage-1 optimum with  $\pm 15\%$  of each parameter's range, enabling rapid local refinement without escaping the vicinity of the global optimum. Both stages use patience-based early stopping (3 generations without improvement) to avoid unnecessary evaluations once the population has converged. L-BFGS-B polishing is explicitly disabled (`polish=False`) to keep the solution within the discrete-compatible parameter space, and population updates are deferred (`updating="deferred"`) so that all fitness evaluations within a generation complete before any individual is replaced, ensuring reproducible parallel-safe behaviour.

Table 9: Differential Evolution configuration.

| Parameter                      | Stage 1 (coarse)   | Stage 2 (fine)     |
|--------------------------------|--------------------|--------------------|
| Strategy                       | best1bin           | best1bin           |
| Population size ( $\times d$ ) | 8 $\times$ 5       | 12 $\times$ 5      |
| Max iterations                 | 8                  | 12                 |
| Mutation ( $F$ )               | 0.5–1.0 (dithered) | 0.5–1.0 (dithered) |
| Crossover ( $CR$ )             | 0.7                | 0.7                |
| Convergence tol.               | 10 <sup>-4</sup>   | 10 <sup>-4</sup>   |
| Proxy forest size              | 20 trees           | 40 trees           |
| Proxy data cap                 | 2,000 samples      | 7,000 samples      |
| Early stopping (patience)      | 3 generations      | 3 generations      |
| Polishing (L-BFGS-B)           | disabled           | disabled           |
| Population updating            | deferred           | deferred           |

## B. Optimised Hyperparameter Values

Table 10: DE-optimised parameters  $\theta^*$  from Fold 1 of MIMIC-IV, applied to all subsequent folds and eICU.

| Parameter            | Symbol     | Optimised Value |
|----------------------|------------|-----------------|
| Lower quantile level | $\alpha_L$ | 0.0555          |
| Upper quantile level | $\alpha_U$ | 0.9384          |
| Fuzzy proportion     | $\eta$     | 0.2869          |
| Max tree depth       | $d_{\max}$ | 6               |
| Min leaf samples     | $n_{\min}$ | 33              |

## C. Extended eICU Cohort Statistics

Table 11: Cohort summary statistics for MIMIC-IV (primary) and eICU (external validation). Feature counts include binary missingness indicators appended for features with &gt; 5% missing rate. Mortality rates are computed after the 4-hour / 30-day admission-length filter.

| Variable                                        | MIMIC-IV                | eICU                   |
|-------------------------------------------------|-------------------------|------------------------|
| Total ICU stays                                 | 82,785                  | 162,136                |
| Unique patients                                 | 63,687                  | 121,567                |
| Hospitals                                       | 1 (BIDMC)               | 208                    |
| In-hospital deaths                              | 8,762                   | 8,621                  |
| Crude mortality, %                              | 10.58%                  | 5.32%                  |
| Raw features                                    | 155                     | 155                    |
| + Missingness indicators                        | +26                     | +74                    |
| Total features                                  | 181                     | 229                    |
| CV folds                                        | 5 (patient-level)       | 5 (patient-level)      |
| <i>Per-fold stays / deaths / mortality rate</i> |                         |                        |
| Fold 1                                          | 16,557 / 1,792 / 10.82% | 32,428 / 1,731 / 5.34% |
| Fold 2                                          | 16,557 / 1,762 / 10.64% | 32,427 / 1,729 / 5.33% |
| Fold 3                                          | 16,557 / 1,750 / 10.57% | 32,427 / 1,731 / 5.34% |
| Fold 4                                          | 16,557 / 1,688 / 10.20% | 32,427 / 1,683 / 5.19% |
| Fold 5                                          | 16,557 / 1,770 / 10.69% | 32,427 / 1,747 / 5.39% |

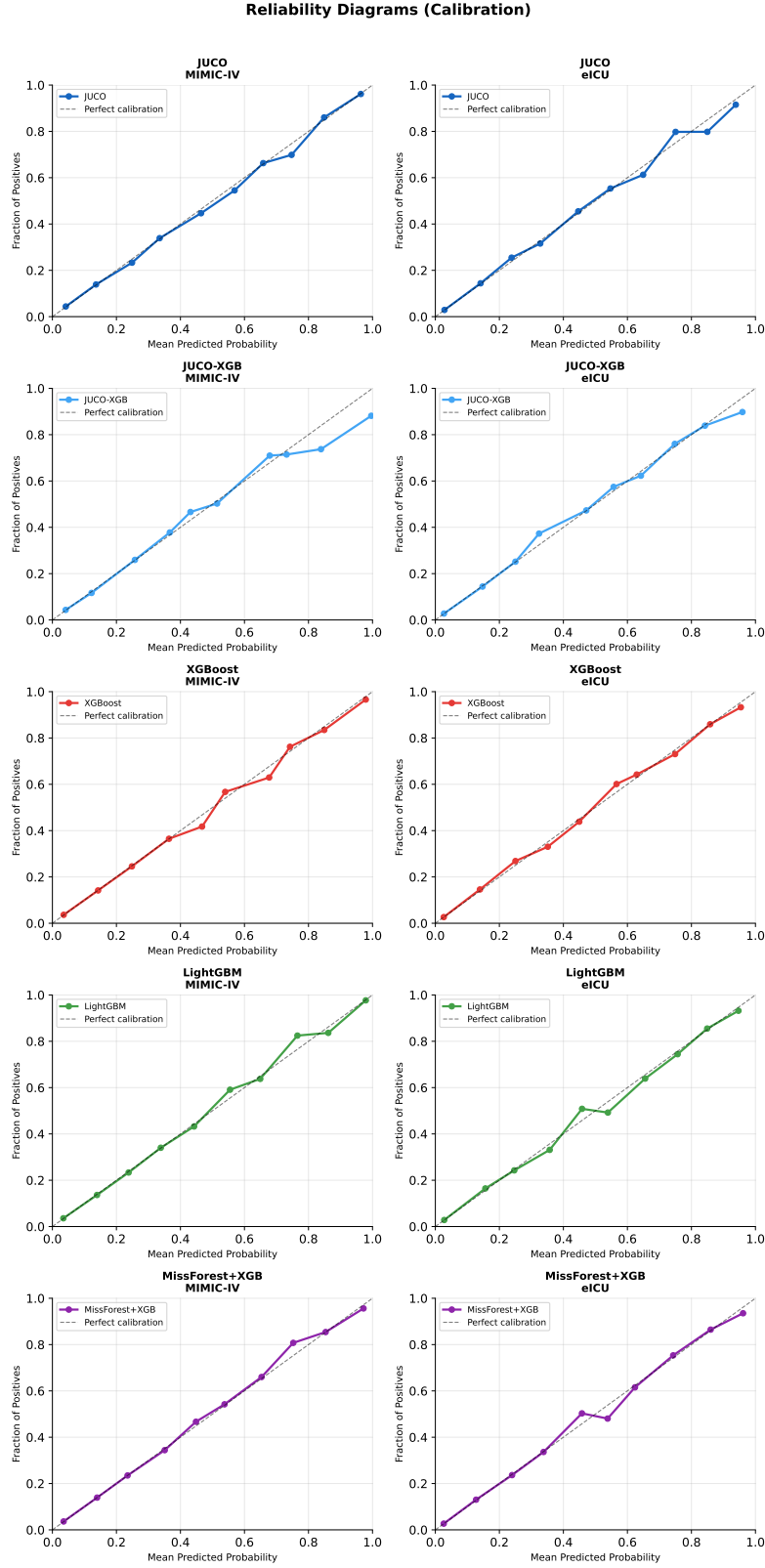


Figure 7: Reliability diagrams for all models on MIMIC-IV (top row) and eICU (bottom row). The diagonal dashed line represents perfect calibration. JUCO achieves  $ECE = 0.0063$  on MIMIC-IV and  $ECE = 0.0029$  on eICU (frozen isotonic calibrator), demonstrating stable calibration under cross-institutional distribution shift.